

Large Language Models

Parameter Efficient Fine Tuning II

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Courtesy: Most of the slides are adopted from the papers by Li Liang 2021 “Prefix-Tuning: Optimizing Continuous Prompts for Generation,” Hu et al 2021, “Intrinsic Dimensionality Explains The Effectiveness of Language Model Fine-tuning” Aghajanyan et al 2020, “LoRA: Low-rank Adaptation Of Large Language Models” and He et al. 2022 “Towards A Unified View of Parameter-efficient Transfer Learning”

Motivation

- Providing proper **task-specific context** in the input can steer the LM to solve the task more efficiently.
- Encoding of the original input x will **change**. Why?
 - Guiding the model to **extract relevant** information from x .
- Does this context **exist**? How to **find** it?

من خواهم از تانک های بهینه سازی پیوسته استفاده کنم
در پرسش token به دردت افزاینده کنم و سعی کنم این token ها
در فضای embedding یاد بگیرم

Prefix Tuning

Prefix-Tuning: Optimizing Continuous Prompts for Generation

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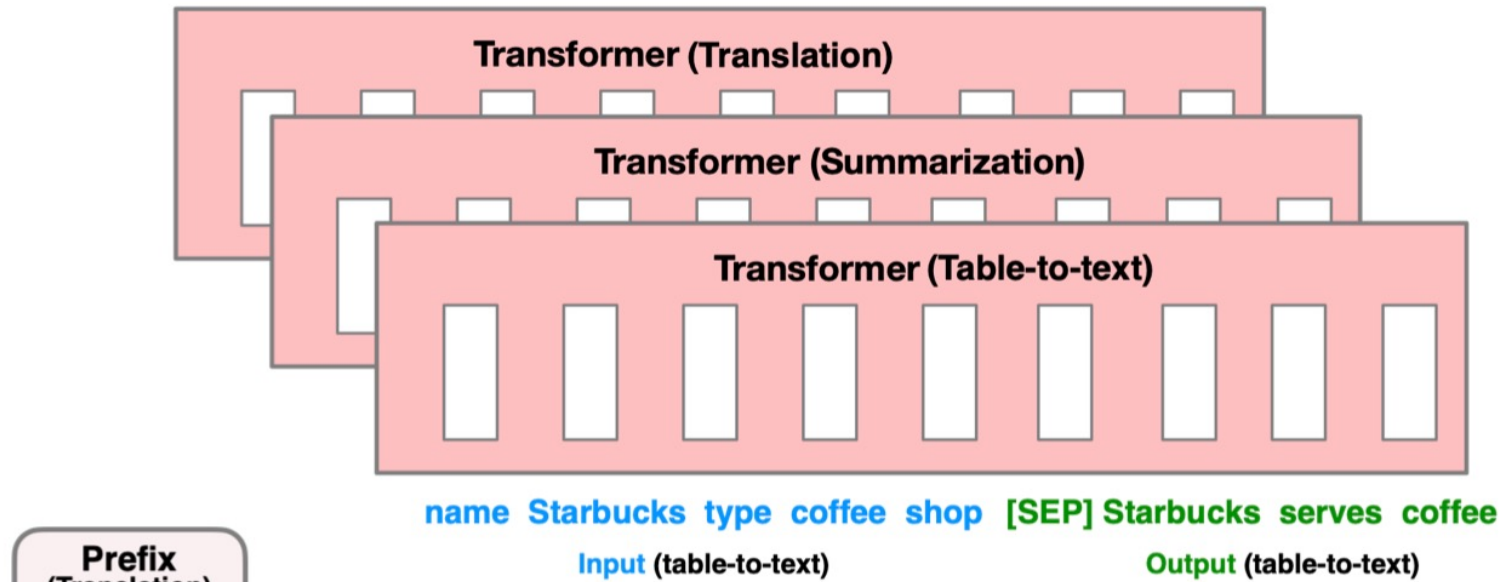
`pliang@cs.stanford.edu`

- Prepend certain **trainable prefix tokens** to the input/hidden activations.
- The **hidden representation** becomes:

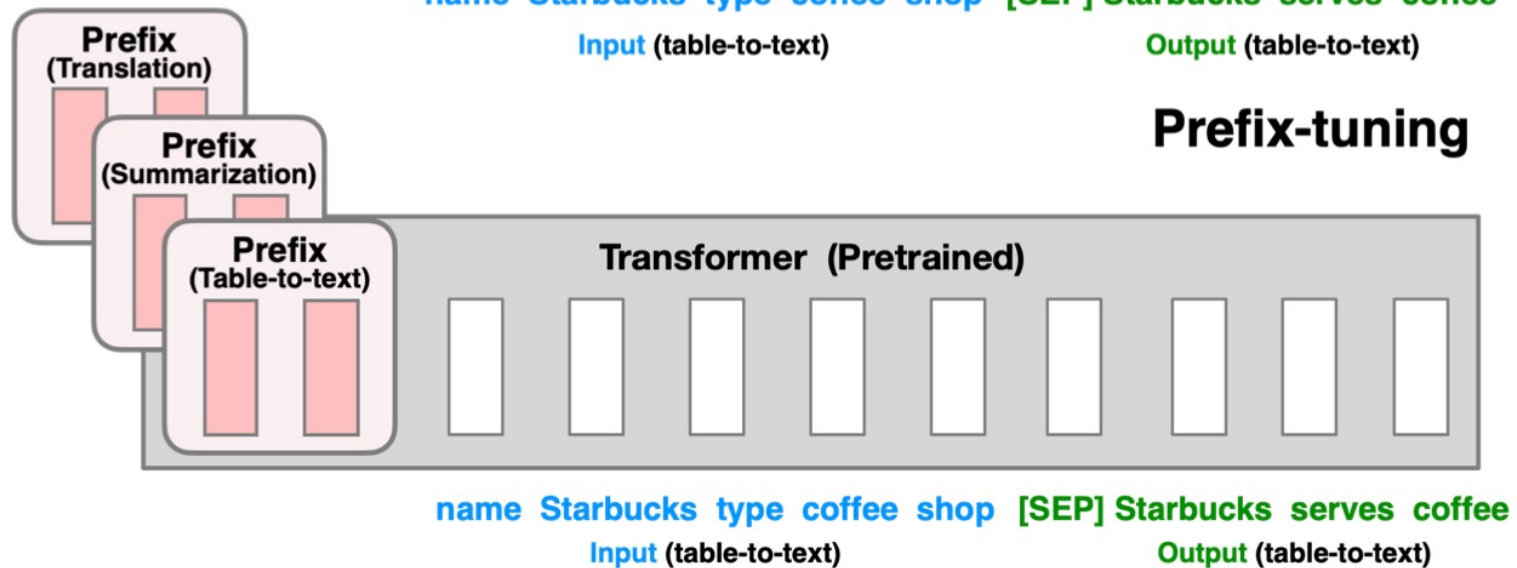
$$h_i = \begin{cases} P_\theta[i, :], & \text{if } i \in P_{\text{idx}}, \\ \text{LM}_\phi(z_i, h_{<i}), & \text{otherwise.} \end{cases}$$

- **All** h_i 's would indeed be a function of the trainable parameters P_θ . Why?

Fine-tuning

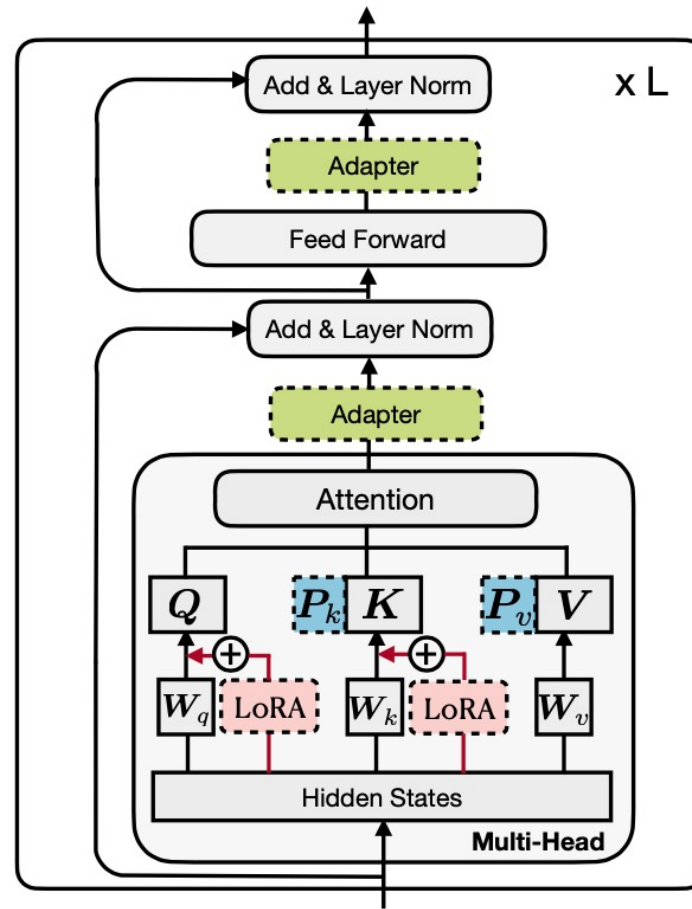


Prefix-tuning

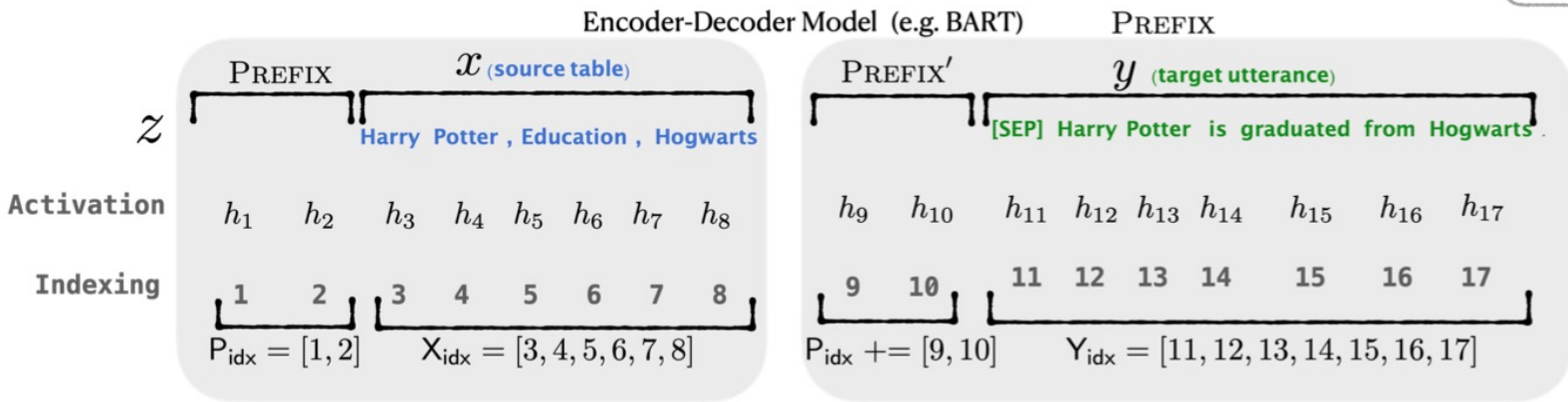
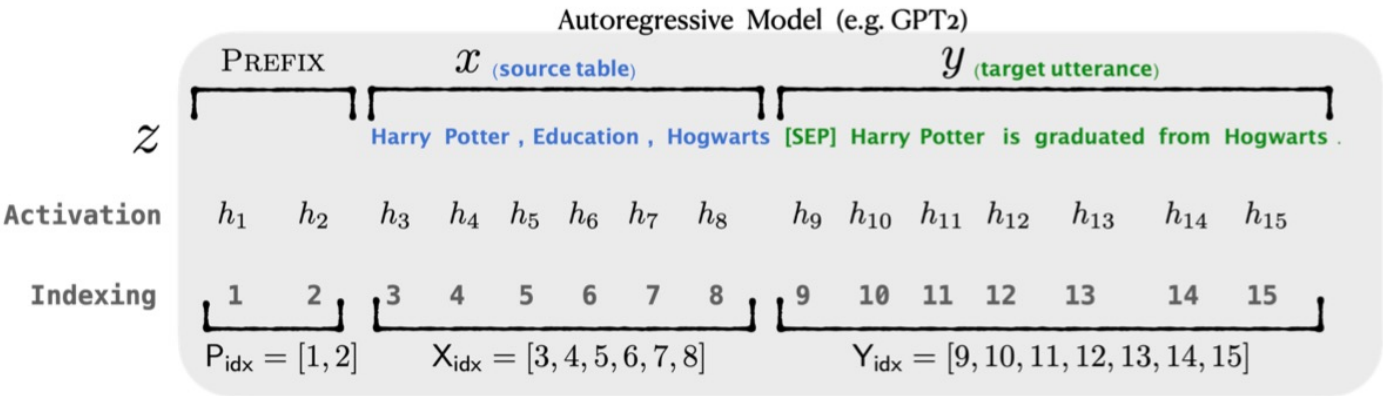


Prefix Tuning (cont.)

$$\text{head}_i = \text{Attn}(\mathbf{x}\mathbf{W}_q^{(i)}, \text{concat}(\mathbf{P}_k^{(i)}, \mathbf{C}\mathbf{W}_k^{(i)}), \text{concat}(\mathbf{P}_v^{(i)}, \mathbf{C}\mathbf{W}_v^{(i)}))$$



Prefix Tuning (cont.)



Summarization Example

Article: Scientists at University College London discovered people tend to think that their hands are wider and their fingers are shorter than they truly are.They say the confusion may lie in the way the brain receives information from different parts of the body.Distorted perception may dominate in some people, leading to body image problems ... [ignoring 308 words] could be very motivating for people with eating disorders to know that there was a biological explanation for their experiences, rather than feeling it was their fault."

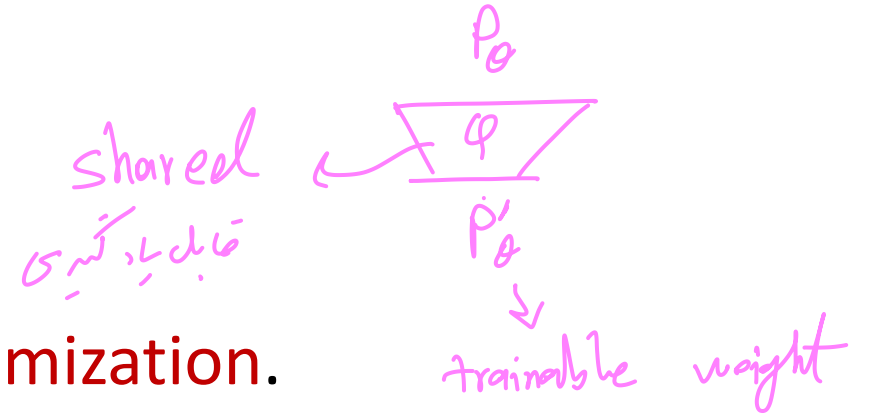
Summary: The brain naturally distorts body image - a finding which could explain eating disorders like anorexia, say experts.

Table-to-text Example

Table: name[Clowns] customer-rating[1 out of 5] eatType[coffee shop] food[Chinese] area[riverside] near[Clare Hall]

Textual Description: Clowns is a coffee shop in the riverside area near Clare Hall that has a rating 1 out of 5 . They serve Chinese food .

Parametrization of P_θ



- Directly optimizing P_θ leads to **unstable optimization**.
 - Slight drop in performance.
- Use a **smaller** P'_θ as input to an MLP with **shared** trainable weights ϕ .
- So $P_\theta = MLP_\phi(P'_\theta)$.
- We can **drop** P'_θ after training and use the result (P_θ).

over fit سے بچانے کے لیے Full-fine-tune

Results

	E2E					WebNLG									DART					
	BLEU	NIST	MET	R-L	CIDEr	BLEU			MET			TER ↓			BLEU	MET	TER ↓	Mover	BERT	BLEURT
						S	U	A	S	U	A	S	U	A						
GPT-2 _{MEDIUM}																				
FINE-TUNE	68.2	8.62	46.2	71.0	2.47	64.2	27.7	46.5	0.45	0.30	0.38	0.33	0.76	0.53	46.2	0.39	0.46	0.50	0.94	0.39
FT-TOP2	68.1	8.59	46.0	70.8	2.41	53.6	18.9	36.0	0.38	0.23	0.31	0.49	0.99	0.72	41.0	0.34	0.56	0.43	0.93	0.21
ADAPTER(3%)	68.9	8.71	46.1	71.3	2.47	60.4	48.3	54.9	0.43	0.38	0.41	0.35	0.45	0.39	45.2	0.38	0.46	0.50	0.94	0.39
ADAPTER(0.1%)	66.3	8.41	45.0	69.8	2.40	54.5	45.1	50.2	0.39	0.36	0.38	0.40	0.46	0.43	42.4	0.36	0.48	0.47	0.94	0.33
PREFIX(0.1%)	69.7	8.81	46.1	71.4	2.49	62.9	45.6	55.1	0.44	0.38	0.41	0.35	0.49	0.41	46.4	0.38	0.46	0.50	0.94	0.39
GPT-2 _{LARGE}																				
FINE-TUNE	68.5	8.78	46.0	69.9	2.45	65.3	43.1	55.5	0.46	0.38	0.42	0.33	0.53	0.42	47.0	0.39	0.46	0.51	0.94	0.40
Prefix	70.3	8.85	46.2	71.7	2.47	63.4	47.7	56.3	0.45	0.39	0.42	0.34	0.48	0.40	46.7	0.39	0.45	0.51	0.94	0.40
SOTA	68.6	8.70	45.3	70.8	2.37	63.9	52.8	57.1	0.46	0.41	0.44	-	-	-	-	-	-	-	-	-

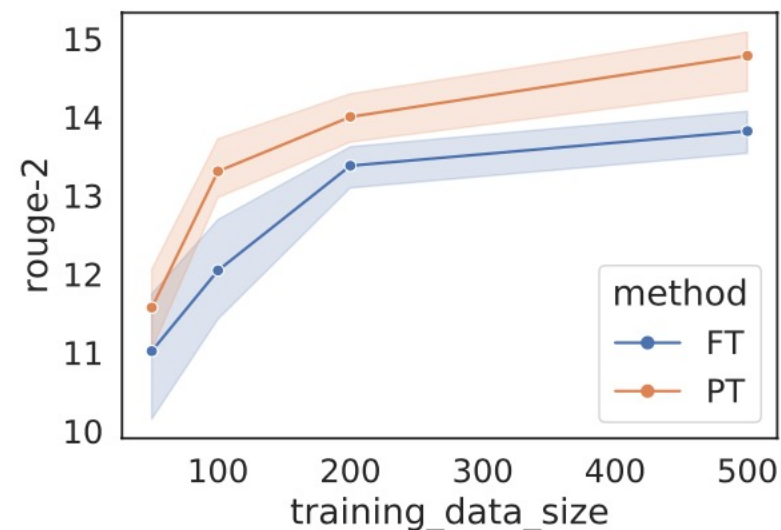
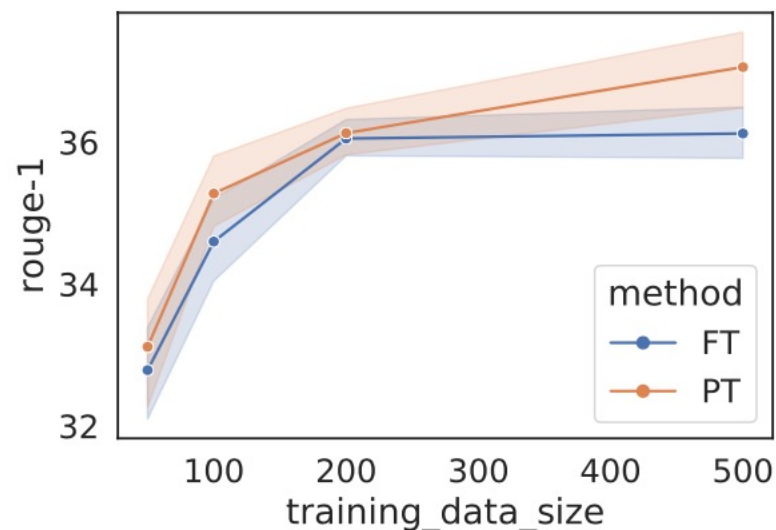
Table 1: Metrics (higher is better, except for TER) for table-to-text generation on E2E (left), WebNLG (middle) and DART (right). With only 0.1% parameters, Prefix-tuning outperforms other lightweight baselines and achieves a comparable performance with fine-tuning. The best score is boldfaced for both GPT-2_{MEDIUM} and GPT-2_{LARGE}.

Qualitative Results on Table-to-Text (low data setting)

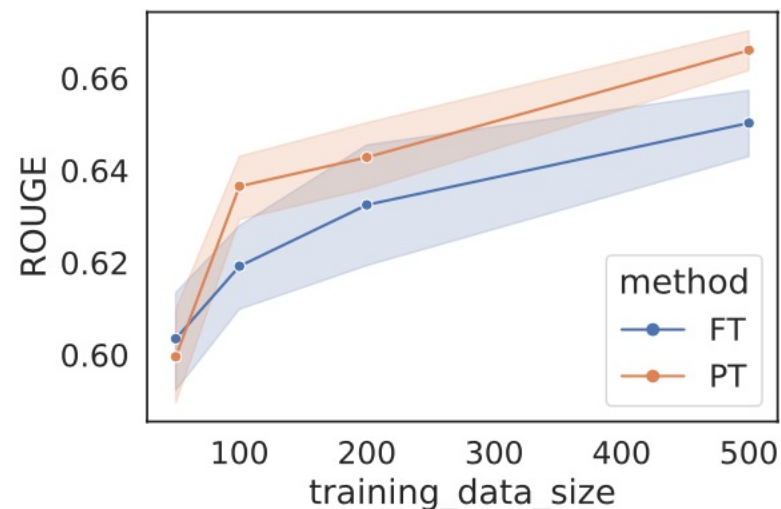
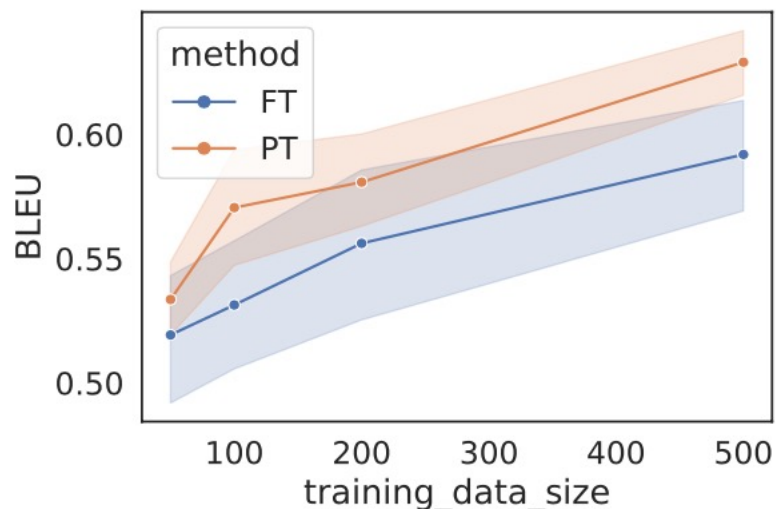
Source	name : The Eagle type : coffee shop food : Chinese price : cheap customer rating : average area : riverside family friendly : no near : Burger King
Prefix (50)	The Eagle is a cheap Chinese coffee shop located near Burger King.
Prefix (100)	The Eagle is a cheap coffee shop located in the riverside near Burger King. It has average customer ratings.
Prefix (200)	The Eagle is a cheap Chinese coffee shop located in the riverside area near Burger King. It has average customer ratings.
Prefix (500)	The Eagle is a coffee shop that serves Chinese food. It is located in the riverside area near Burger King. It has an average customer rating and is not family friendly.
FT (50)	The Eagle coffee shop is located in the riverside area near Burger King.
FT (100)	The Eagle is a cheap coffee shop near Burger King in the riverside area. It has a low customer rating and is not family friendly.
FT (200)	The Eagle is a cheap Chinese coffee shop with a low customer rating. It is located near Burger King in the riverside area.
FT (500)	The Eagle is a cheap Chinese coffee shop with average customer ratings. It is located in the riverside area near Burger King.

Prefix-tuning Outperforms FT in low-data regimes

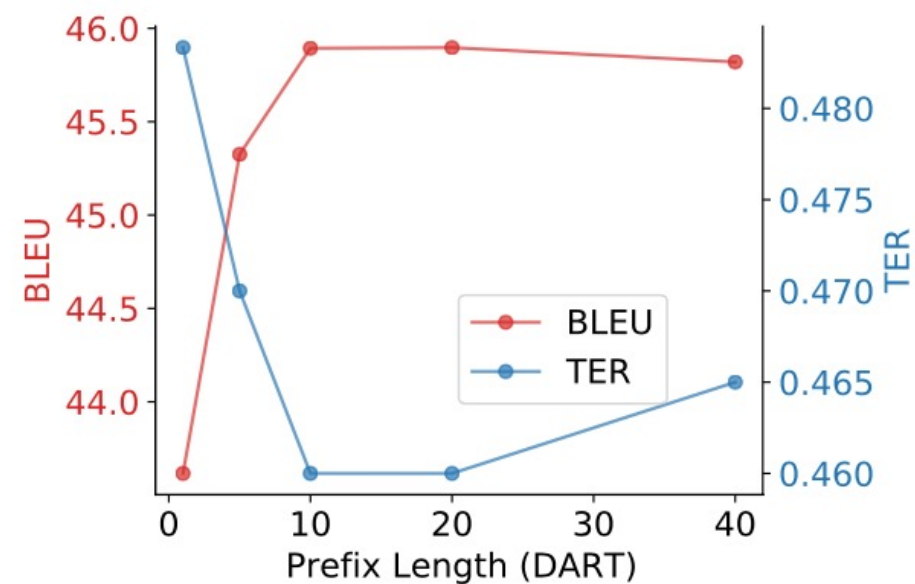
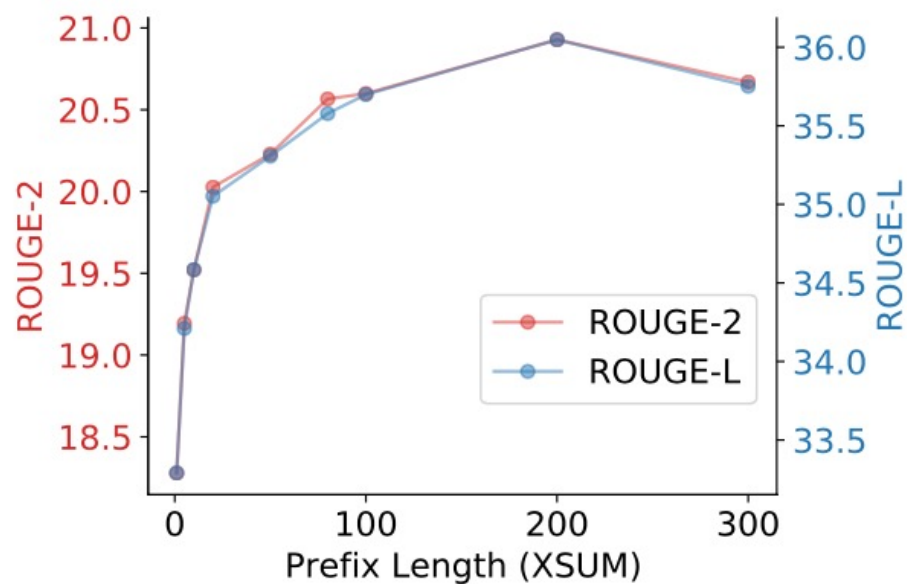
Summarization



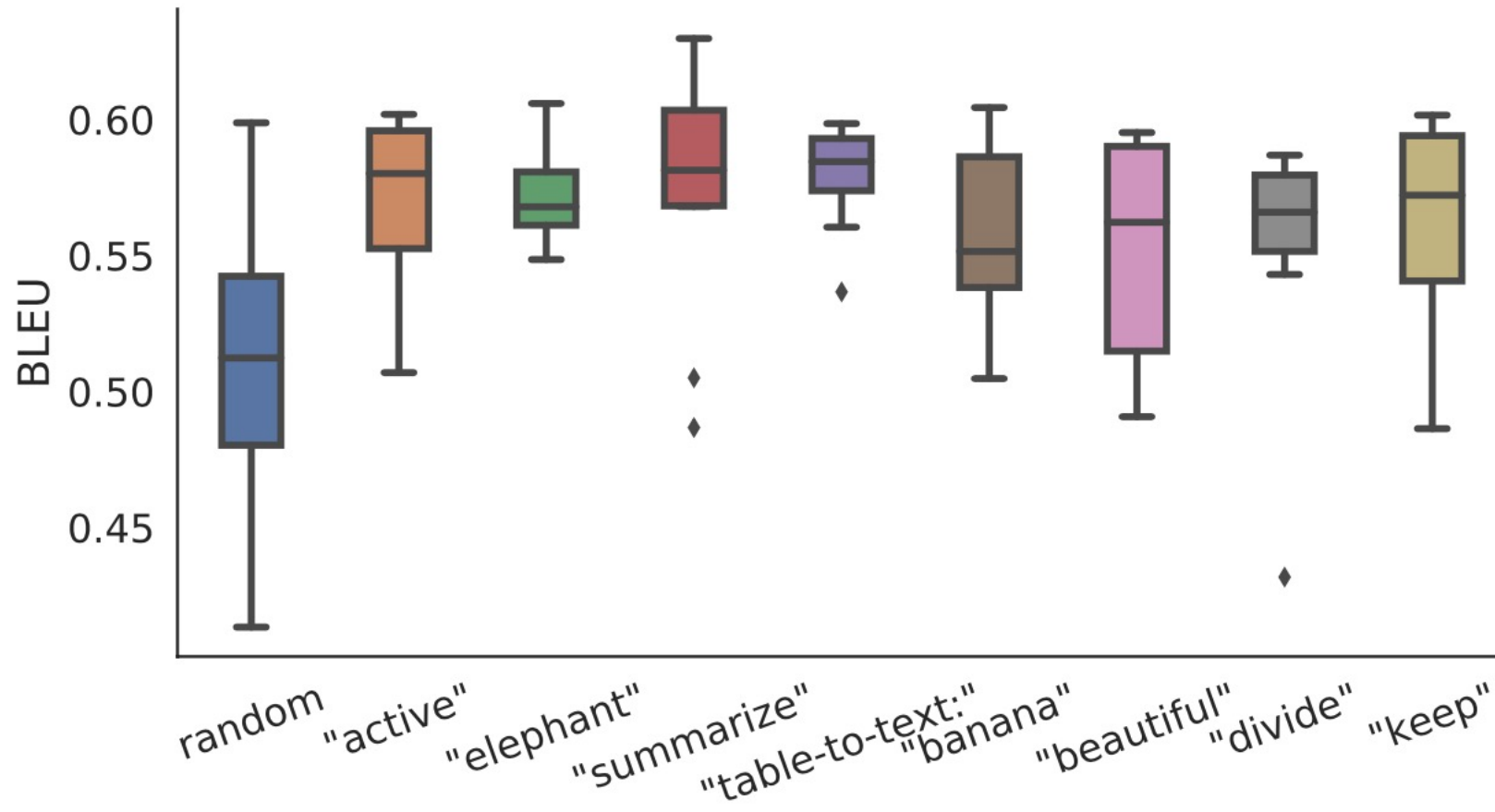
Text-to-Table



Ablation (Prefix length)



Ablation (Initialization of Prefixes)



INTRINSIC DIMENSIONALITY EXPLAINS THE EFFECTIVENESS OF LANGUAGE MODEL FINE-TUNING

Armen Aghajanyan, Luke Zettlemoyer, Sonal Gupta

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Intrinsic Dimensionality of a Model

- Model with trainable parameters $\theta^D \in \mathbb{R}^D$.
- Map θ to a **lower dimensional** space $\theta^d \in \mathbb{R}^d$.
- Solve the optimization (training) in that space:

$$\theta^D = \theta_0^D + P(\theta^d)$$

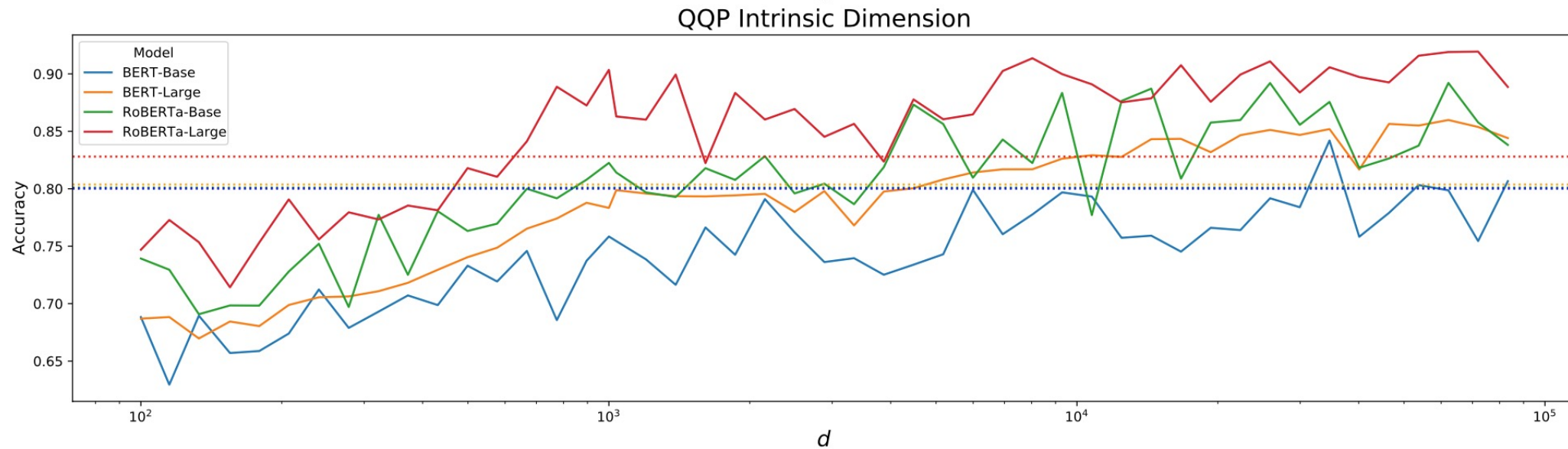
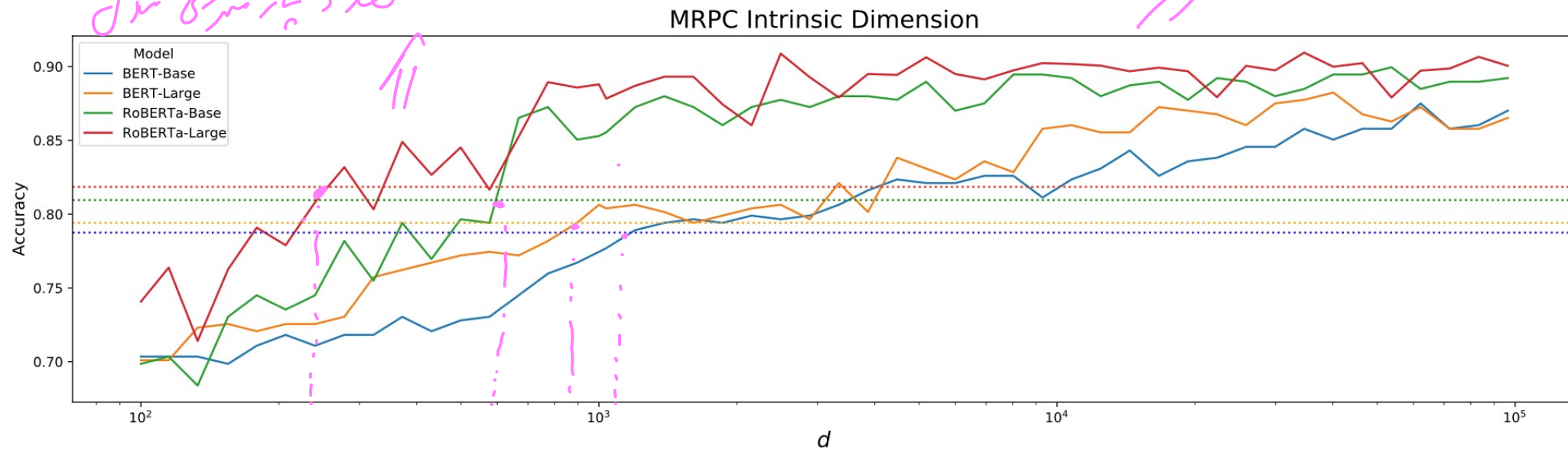
with $\theta^D = P(\theta^d)$ (FastFood Transform)

- Let **d**₉₀ be the dimensionality that results to 90% of the performance of full fine tuning.
- Structure aware intrinsic dimension $\theta_i^D = \theta_{0,i}^D + \lambda_i P(\theta^{d-m})_i$

!!!!

بعد ازاتی مدل α
تعداد پارامترهای مدل

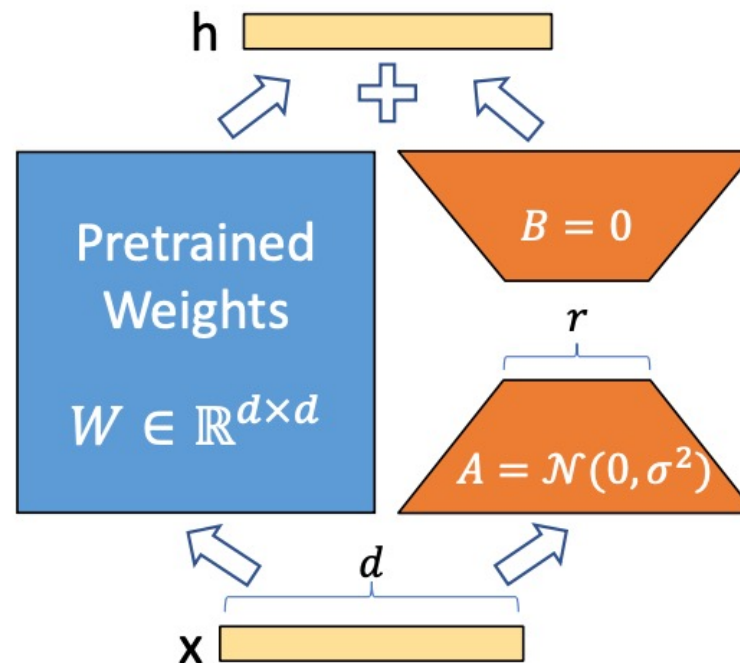
بعد ازاتی مدل های α



LoRA (Low Rank Adaptation)

این مدل با رتبه پائین

- Learned overparameterized models facilitate learning on a **low dimensional space**.
- So ... weight updates could possibly be **low rank**.



LoRA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS

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Yuanzhi Li **Shean Wang** **Lu Wang** **Weizhu Chen**

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(Version 2)

Problem Statement

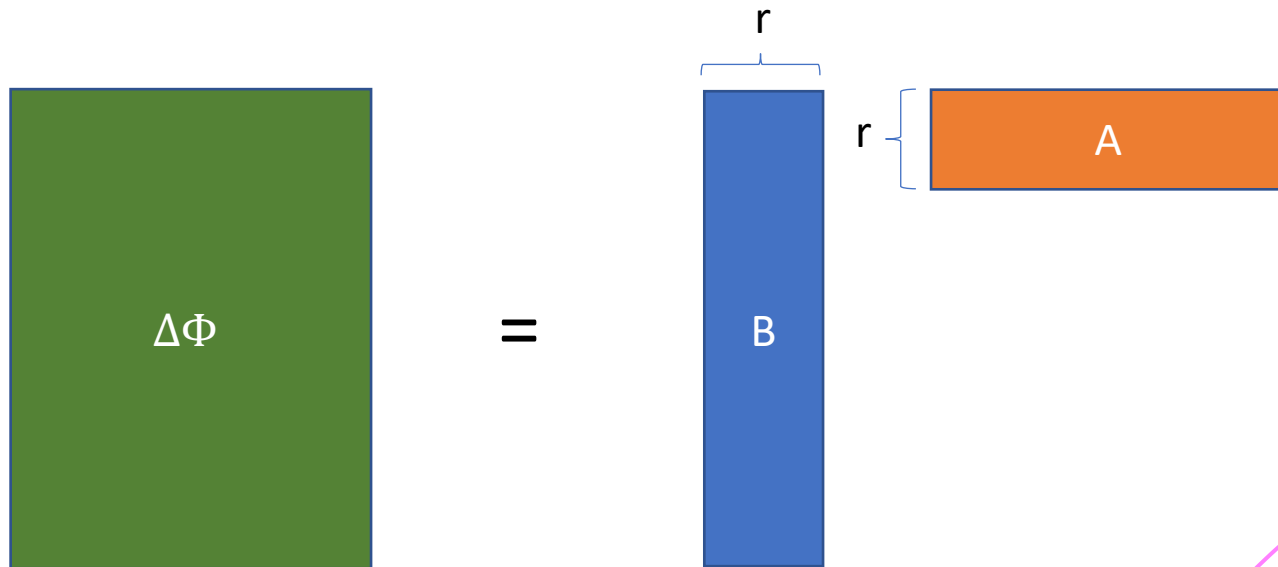
- Given a pretrained **autoregressive** language model $P_{\Phi_0}(y|x)$.
- Also given a downstream **conditional** text generation task $\mathcal{Z} = \{(x_i, y_i)\}_{i=1..N}$.
 - e.g. NL2SQL $x_i = \text{seq. of natural lang. query}$; $y_i = \text{SQL command}$
- Update the weights to $\Phi_0 + \Delta\Phi$ to optimize:

$$\max_{\Phi} \sum_{(x,y) \in \mathcal{Z}} \sum_{t=1}^{|y|} \log(P_{\Phi}(y_t|x, y_{<t}))$$

- Now let $\Delta\Phi(\Theta)$ be a function of Θ , which lives in a **lower dimensional** space.

Solution

- We let $\Delta\Phi(\Theta) = BA$, so $\Theta = (A, B)$.



از حالت تابع
محاسبات شروع
میکنیم
← اینها را میزنیم

- Random **Gaussian** initialization of A , and $B = 0$. Why?
- Only weights in the **self-attention module are trainable**; MLPs are frozen.

Results

Model & Method	# Trainable Parameters	MNLI	SST-2	MRPC	CoLA	QNLI	QQP	RTE	STS-B	Avg.
RoB _{base} (FT)*	125.0M	87.6	94.8	90.2	63.6	92.8	91.9	78.7	91.2	86.4
RoB _{base} (BitFit)*	0.1M	84.7	93.7	92.7	62.0	91.8	84.0	81.5	90.8	85.2
RoB _{base} (Adpt ^D)*	0.3M	87.1 $\pm$.0	94.2 $\pm$.1	88.5 $\pm$ 1.1	60.8 $\pm$.4	93.1 $\pm$.1	90.2 $\pm$.0	71.5 $\pm$ 2.7	89.7 $\pm$.3	84.4
RoB _{base} (Adpt ^D)*	0.9M	87.3 $\pm$.1	94.7 $\pm$.3	88.4 $\pm$.1	62.6 $\pm$.9	93.0 $\pm$.2	90.6 $\pm$.0	75.9 $\pm$ 2.2	90.3 $\pm$.1	85.4
RoB _{base} (LoRA)	0.3M	87.5 $\pm$.3	95.1 $\pm$.2	89.7 $\pm$.7	63.4 $\pm$ 1.2	93.3 $\pm$.3	90.8 $\pm$.1	86.6 $\pm$.7	91.5 $\pm$.2	87.2
RoB _{large} (FT)*	355.0M	90.2	96.4	90.9	68.0	94.7	92.2	86.6	92.4	88.9
RoB _{large} (LoRA)	0.8M	90.6 $\pm$.2	96.2 $\pm$.5	90.9 $\pm$ 1.2	68.2 $\pm$ 1.9	94.9 $\pm$.3	91.6 $\pm$.1	87.4 $\pm$ 2.5	92.6 $\pm$.2	89.0
RoB _{large} (Adpt ^P)†	3.0M	90.2 $\pm$.3	96.1 $\pm$.3	90.2 $\pm$.7	68.3 $\pm$ 1.0	94.8 $\pm$.2	91.9 $\pm$.1	83.8 $\pm$ 2.9	92.1 $\pm$.7	88.4
RoB _{large} (Adpt ^P)†	0.8M	90.5 $\pm$.3	96.6 $\pm$.2	89.7 $\pm$ 1.2	67.8 $\pm$ 2.5	94.8 $\pm$.3	91.7 $\pm$.2	80.1 $\pm$ 2.9	91.9 $\pm$.4	87.9
RoB _{large} (Adpt ^H)†	6.0M	89.9 $\pm$.5	96.2 $\pm$.3	88.7 $\pm$ 2.9	66.5 $\pm$ 4.4	94.7 $\pm$.2	92.1 $\pm$.1	83.4 $\pm$ 1.1	91.0 $\pm$ 1.7	87.8
RoB _{large} (Adpt ^H)†	0.8M	90.3 $\pm$.3	96.3 $\pm$.5	87.7 $\pm$ 1.7	66.3 $\pm$ 2.0	94.7 $\pm$.2	91.5 $\pm$.1	72.9 $\pm$ 2.9	91.5 $\pm$.5	86.4
RoB _{large} (LoRA)†	0.8M	90.6 $\pm$.2	96.2 $\pm$.5	90.2 $\pm$ 1.0	68.2 $\pm$ 1.9	94.8 $\pm$.3	91.6 $\pm$.2	85.2 $\pm$ 1.1	92.3 $\pm$.5	88.6
DeB _{XXL} (FT)*	1500.0M	91.8	97.2	92.0	72.0	96.0	92.7	93.9	92.9	91.1
DeB _{XXL} (LoRA)	4.7M	91.9 $\pm$.2	96.9 $\pm$.2	92.6 $\pm$.6	72.4 $\pm$ 1.1	96.0 $\pm$.1	92.9 $\pm$.1	94.9 $\pm$.4	93.0 $\pm$.2	91.3

Table 2: RoBERTa_{base}, RoBERTa_{large}, and DeBERTa_{XXL} with different adaptation methods on the GLUE benchmark. We report the overall (matched and mismatched) accuracy for MNLI, Matthew’s correlation for CoLA, Pearson correlation for STS-B, and accuracy for other tasks. Higher is better for all metrics. * indicates numbers published in prior works. † indicates runs configured in a setup similar to [Houlsby et al. \(2019\)](#) for a fair comparison.

Results (cont.)

Model & Method	# Trainable Parameters	E2E NLG Challenge				
		BLEU	NIST	MET	ROUGE-L	CIDEr
GPT-2 M (FT)*	354.92M	68.2	8.62	46.2	71.0	2.47
GPT-2 M (Adapter ^L)*	0.37M	66.3	8.41	45.0	69.8	2.40
GPT-2 M (Adapter ^L)*	11.09M	68.9	8.71	46.1	71.3	2.47
GPT-2 M (Adapter ^H)	11.09M	67.3 $\pm$.6	8.50 $\pm$.07	46.0 $\pm$.2	70.7 $\pm$.2	2.44 $\pm$.01
GPT-2 M (FT ^{Top2})*	25.19M	68.1	8.59	46.0	70.8	2.41
GPT-2 M (PreLayer)*	0.35M	69.7	8.81	46.1	71.4	2.49
GPT-2 M (LoRA)	0.35M	70.4$\pm$.1	8.85$\pm$.02	46.8$\pm$.2	71.8$\pm$.1	2.53$\pm$.02
GPT-2 L (FT)*	774.03M	68.5	8.78	46.0	69.9	2.45
GPT-2 L (Adapter ^L)	0.88M	69.1 $\pm$.1	8.68 $\pm$.03	46.3 $\pm$.0	71.4 $\pm$.2	2.49$\pm$.0
GPT-2 L (Adapter ^L)	23.00M	68.9 $\pm$.3	8.70 $\pm$.04	46.1 $\pm$.1	71.3 $\pm$.2	2.45 $\pm$.02
GPT-2 L (PreLayer)*	0.77M	70.3	8.85	46.2	71.7	2.47
GPT-2 L (LoRA)	0.77M	70.4$\pm$.1	8.89$\pm$.02	46.8$\pm$.2	72.0$\pm$.2	2.47 $\pm$.02

Table 3: GPT-2 medium (M) and large (L) with different adaptation methods on the E2E NLG Challenge. For all metrics, higher is better. LoRA outperforms several baselines with comparable or fewer trainable parameters. Confidence intervals are shown for experiments we ran. * indicates numbers published in prior works.

Results (cont.)

Model&Method	# Trainable Parameters	WikiSQL	MNLI-m	SAMSum
		Acc. (%)	Acc. (%)	R1/R2/RL
GPT-3 (FT)	175,255.8M	73.8	89.5	52.0/28.0/44.5
GPT-3 (BitFit)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 (PreEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 (PreLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 (Adapter ^H)	7.1M	71.9	89.8	53.0/28.9/44.8
GPT-3 (Adapter ^H)	40.1M	73.2	91.5	53.2/29.0/45.1
GPT-3 (LoRA)	4.7M	73.4	91.7	53.8/29.8/45.9
GPT-3 (LoRA)	37.7M	74.0	91.6	53.4/29.2/45.1

Table 4: Performance of different adaptation methods on GPT-3 175B. We report the logical form validation accuracy on WikiSQL, validation accuracy on MultiNLI-matched, and Rouge-1/2/L on SAMSum. LoRA performs better than prior approaches, including full fine-tuning. The results on WikiSQL have a fluctuation around $\pm 0.5\%$, MNLI-m around $\pm 0.1\%$, and SAMSum around $\pm 0.2/\pm 0.2/\pm 0.1$ for the three metrics.

Comparison to other PEFTs

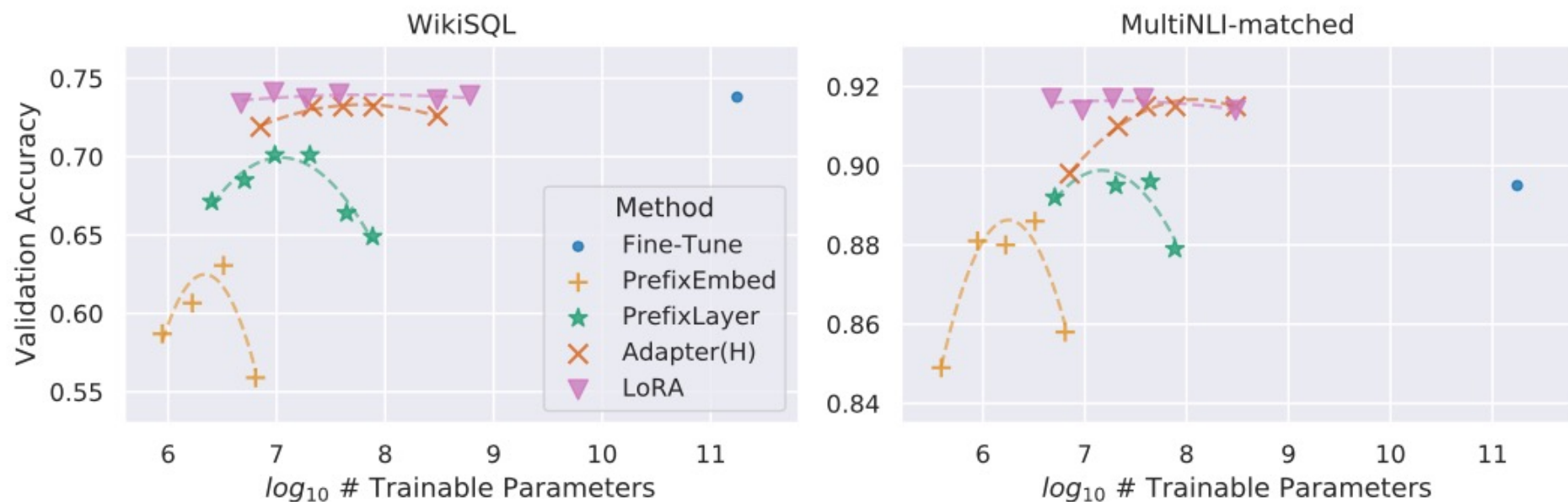


Figure 2: GPT-3 175B validation accuracy vs. number of trainable parameters of several adaptation methods on WikiSQL and MNLI-matched. LoRA exhibits better scalability and task performance. See [Section F.2](#) for more details on the plotted data points.

Why $r=1$ works well in practice?

چرا $r=1$!
هم چیز!

- Let $A_{r=8}$ and $A_{r=64}$ be the **learned matrices** for $r = 8$, and 64.
- Do they extract **similar features** from the token embeddings?
- How to **measure** this?
- Each A can be considered as a **subspace**.
- Find how similar these two subspaces are?

Why $r=1$ works well in practice? (cont.)

$$A = V\Sigma U$$

$$\Rightarrow A = \sum_{i=1}^r \sigma_i v_i u_i^T$$

$$\Rightarrow Ax = \sum_{i=1}^r \sigma_i v_i u_i^T x$$

$$\Rightarrow Ax = \sum_{i=1}^r \sigma_i \langle u_i, x \rangle v_i$$

Pick **highest σ_i** , compare the **corresponding u_i 's** in two A's

Why $r=1$ works well in practice? (cont.)

- Grassmann distance:

$$\phi(A_{r=8}, A_{r=64}, i, j) = \frac{\|U_{A_{r=8}}^{i\top} U_{A_{r=64}}^j\|_F^2}{\min(i, j)} \in [0, 1]$$

Why $r=1$ works well in practice? (cont.)

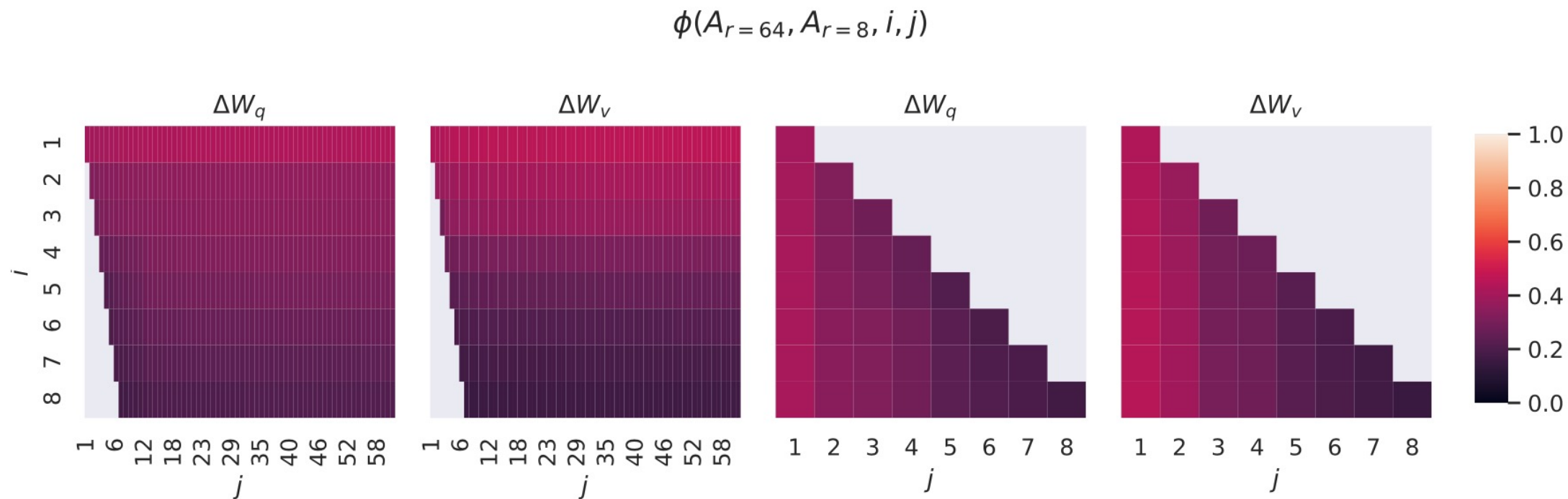


Figure 3: Subspace similarity between column vectors of $A_{r=8}$ and $A_{r=64}$ for both ΔW_q and ΔW_v . The third and the fourth figures zoom in on the lower-left triangle in the first two figures. The top directions in $r=8$ are included in $r=64$, and vice versa.

ΔW only amplifies directions that are **not** emphasized in W

ایا این تغییر ای بای استخراج کردن؟! 

	$r = 4$			$r = 64$		
	ΔW_q	W_q	Random	ΔW_q	W_q	Random
$\ U^\top W_q V^\top\ _F =$	0.32	21.67	0.02	1.90	37.71	0.33
$\ W_q\ _F = 61.95$	$\ \Delta W_q\ _F = 6.91$			$\ \Delta W_q\ _F = 3.57$		

Table 7: The Frobenius norm of $U^\top W_q V^\top$ where U and V are the left/right top r singular vector directions of either (1) ΔW_q , (2) W_q , or (3) a random matrix. The weight matrices are taken from the 48th layer of GPT-3.

ولی جواب نه = ای، نه بلکه این تغییر تا کد بیشتری می کنه که ندی. W. غرض

Unified View of PEFT methods (cont.)

- Adapter

$$\mathbf{h} \leftarrow \mathbf{h} + f(\mathbf{h}\mathbf{W}_{\text{down}})\mathbf{W}_{\text{up}}$$

Unified View of PEFT methods (cont.)

- Prefix tuning

$$\begin{aligned}
 \text{head} &= \text{Attn}(\mathbf{x}\mathbf{W}_q, \text{concat}(\mathbf{P}_k, \mathbf{C}\mathbf{W}_k), \text{concat}(\mathbf{P}_v, \mathbf{C}\mathbf{W}_v)) \\
 &= \text{softmax}(\mathbf{x}\mathbf{W}_q \text{concat}(\mathbf{P}_k, \mathbf{C}\mathbf{W}_k)^\top) \begin{bmatrix} \mathbf{P}_v \\ \mathbf{C}\mathbf{W}_v \end{bmatrix} \\
 &= (1 - \lambda(\mathbf{x})) \text{softmax}(\mathbf{x}\mathbf{W}_q \mathbf{W}_k^\top \mathbf{C}^\top) \mathbf{C}\mathbf{W}_v + \lambda(\mathbf{x}) \text{softmax}(\mathbf{x}\mathbf{W}_q \mathbf{P}_k^\top) \mathbf{P}_v \\
 &= (1 - \lambda(\mathbf{x})) \underbrace{\text{Attn}(\mathbf{x}\mathbf{W}_q, \mathbf{C}\mathbf{W}_k, \mathbf{C}\mathbf{W}_v)}_{\text{standard attention}} + \lambda(\mathbf{x}) \underbrace{\text{Attn}(\mathbf{x}\mathbf{W}_q, \mathbf{P}_k, \mathbf{P}_v)}_{\text{independent of } \mathbf{C}}
 \end{aligned}$$

$$\lambda(\mathbf{x}) = \frac{\sum_i \exp(\mathbf{x}\mathbf{W}_q \mathbf{P}_k^\top)_i}{\sum_i \exp(\mathbf{x}\mathbf{W}_q \mathbf{P}_k^\top)_i + \sum_j \exp(\mathbf{x}\mathbf{W}_q \mathbf{W}_k^\top \mathbf{C}^\top)_j}$$

$$\mathbf{h} \leftarrow (1 - \lambda(\mathbf{x}))\mathbf{h} + \lambda(\mathbf{x})\Delta\mathbf{h}, \quad \Delta\mathbf{h} := \text{softmax}(\mathbf{x}\mathbf{W}_q \mathbf{P}_k^\top) \mathbf{P}_v$$

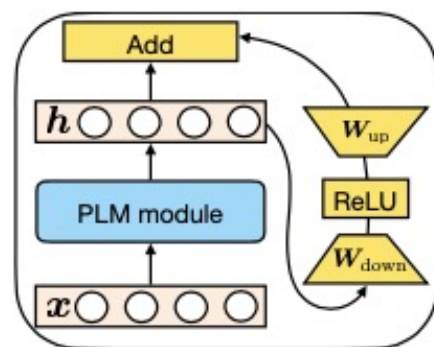
$$\mathbf{h} \leftarrow (1 - \lambda(\mathbf{x}))\mathbf{h} + \lambda(\mathbf{x})f(\mathbf{x}\mathbf{W}_1)\mathbf{W}_2$$

Unified View of PEFT methods (cont.)

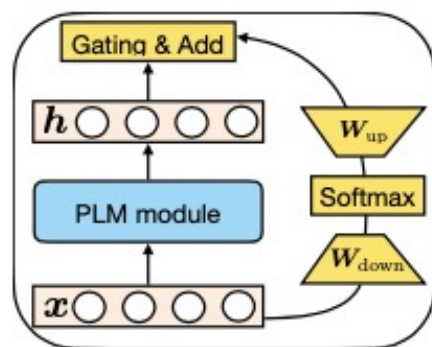
Table 1: Parameter-efficient tuning methods decomposed along the defined design dimensions. Here, for clarity, we directly write the adapter nonlinear function as ReLU which is commonly used. The bottom part of the table exemplifies new variants by transferring design choices of existing approaches.

Method	Δh functional form	insertion form	modified representation	composition function
Existing Methods				
Prefix Tuning	$\text{softmax}(\mathbf{x} \mathbf{W}_q \mathbf{P}_k^\top) \mathbf{P}_v$	parallel	head attn	$\mathbf{h} \leftarrow (1 - \lambda) \mathbf{h} + \lambda \Delta \mathbf{h}$
Adapter	$\text{ReLU}(\mathbf{h} \mathbf{W}_{\text{down}}) \mathbf{W}_{\text{up}}$	sequential	ffn/attn	$\mathbf{h} \leftarrow \mathbf{h} + \Delta \mathbf{h}$
LoRA	$\mathbf{x} \mathbf{W}_{\text{down}} \mathbf{W}_{\text{up}}$	parallel	attn key/val	$\mathbf{h} \leftarrow \mathbf{h} + s \cdot \Delta \mathbf{h}$
Proposed Variants				
Parallel adapter	$\text{ReLU}(\mathbf{h} \mathbf{W}_{\text{down}}) \mathbf{W}_{\text{up}}$	parallel	ffn/attn	$\mathbf{h} \leftarrow \mathbf{h} + \Delta \mathbf{h}$
Muti-head parallel adapter	$\text{ReLU}(\mathbf{h} \mathbf{W}_{\text{down}}) \mathbf{W}_{\text{up}}$	parallel	head attn	$\mathbf{h} \leftarrow \mathbf{h} + \Delta \mathbf{h}$
Scaled parallel adapter	$\text{ReLU}(\mathbf{h} \mathbf{W}_{\text{down}}) \mathbf{W}_{\text{up}}$	parallel	ffn/attn	$\mathbf{h} \leftarrow \mathbf{h} + s \cdot \Delta \mathbf{h}$

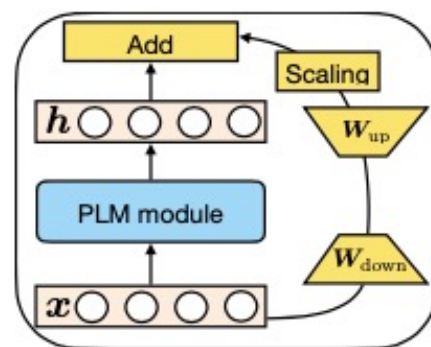
Unified View of PEFT methods (cont.)



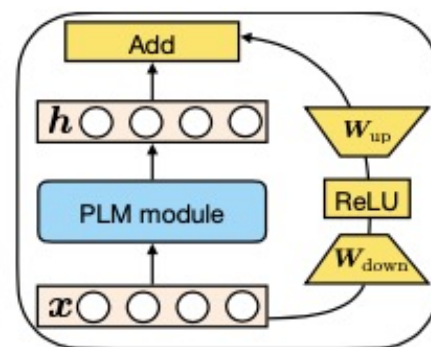
(a) Adapter



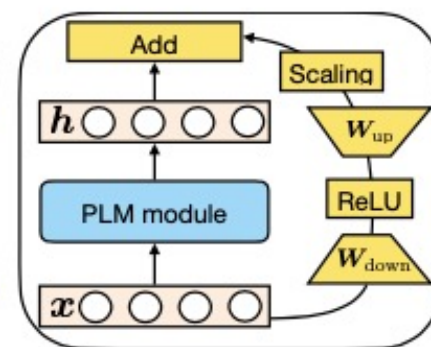
(b) Prefix Tuning



(c) LoRA



(d) Parallel Adapter



(e) Scaled PA

Remarks

- Prefix tuning can be thought of as a “parallel” computation to the PLM layer, whereas the typical adapter is “sequential” computation.
- Adapters are more flexible w.r.t. where they are inserted than prefix tuning
 - Adapters typically modify attention or FFN outputs, while prefix tuning only modifies the attention output of each head.
- Prefix tuning applies to each attention head, while adapters are always single-headed, which makes prefix tuning more expressive.